Molar Concentration - Answer key

GCE Advanced Level Chemistry - Concentration molaire - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. Exercise 1 - From grams to moles

Calculate the number of moles in 10 g of calcium carbonate, CaCO3 (M = 100 g/mol).
Calcule le nombre de moles dans 10 g de carbonate de calcium, CaCO3 (M = 100 g/mol).

1. Corrige / Answer

Ex. 1 - n = m / M = 10 / 100 = 0,10 mol.

2. Exercise 2 - Concentration from moles

0.25 mol of solute is dissolved to make 0.50 L of solution. Compute the molar concentration.
On dissout 0,25 mol de solute pour obtenir 0,50 L de solution. Calcule la concentration molaire.

2. Corrige / Answer

Ex. 2 - c = n / V = 0,25 / 0,50 = 0,50 mol/L.

3. Exercise 3 - The full chain

5.85 g of sodium chloride, NaCl (M = 58.5 g/mol), are made up to 0.50 L of solution. Find the molarity. (First grams to moles, then divide.)
On dissout 5,85 g de chlorure de sodium, NaCl (M = 58.5 g/mol), pour obtenir 0,50 L de solution. Trouve la molarite. (D'abord les grammes en moles, puis on divise.)

3. Corrige / Answer

Ex. 3 - n = m / M = 5,85 / 58,5 = 0,10 mol. c = n / V = 0,10 / 0,50 = 0,20 mol/L.

4. Exercise 4 - Dilution keeps the moles

10 mL of a 2 mol/L stock solution is diluted to 100 mL. What is the final concentration?
On dilue 10 mL d'une solution mere a 2 mol/L jusqu'a 100 mL. Quelle est la concentration finale ?

4. Corrige / Answer

Ex. 4 - C1 V1 = C2 V2 : 2 x 10 = C2 x 100, d'ou C2 = 0,20 mol/L.

5. Exercise 5 - How much mass to weigh

What mass of glucose (M = 180 g/mol) is needed to prepare 0.250 L of a 0.20 mol/L glucose solution?
Quelle masse de glucose (M = 180 g/mol) faut-il peser pour preparer 0,250 L d'une solution de glucose a 0,20 mol/L ?

5. Corrige / Answer

Ex. 5 - n = c V = 0,20 x 0,250 = 0,050 mol. m = n M = 0,050 x 180 = 9,0 g.

6. Exercise 6 - Read the bottle

A bottle is labelled 0.50 mol/L. How many moles of solute are there in 250 mL taken from it? Watch the volume unit.
Un flacon porte l'etiquette 0,50 mol/L. Combien de moles de solute y a-t-il dans 250 mL preleves ? Attention a l'unite de volume.

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6. Corrige / Answer

Ex. 6 - V = 250 mL = 0,250 L. n = c V = 0,50 x 0,250 = 0,125 mol.

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